

Continuous Mappings

5.1 Continuous Mappings on Topological Spaces

Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be two topological spaces.

5.1.1 DEFINITION. A function $f : X \rightarrow Y$ is said to be **continuous at a point $x \in X$** if for each neighborhood $U \in \mathcal{V}_{f(x)}$ there exists a neighborhood $V \in \mathcal{V}_x$ such that

$$f(V) \subseteq U.$$

5.1.2 THEOREM. A function $f : X \rightarrow Y$ is **continuous on X** iff the inverse image under f of every open (closed) set in $(Y, \mathcal{T}_Y)$ is an open (closed) set in $(X, \mathcal{T}_X)$.

5.1.3 REMARK. In general, the image of an open (closed) set under a continuous function is not an open (closed) set.

5.1.4 THEOREM. If $f : (X, \mathcal{T}_X) \rightarrow (Y, \mathcal{T}_Y)$ is a continuous mapping and $A \subseteq X$ is a compact set, then the set $f(A)$ is compact.

👉 no proof!

5.1.5 THEOREM. If $f : (X, \mathcal{T}_X) \rightarrow \mathbb{R}$ is a continuous mapping and $A \subseteq X$ is a compact set, then f is bounded on A .

Proof. Since A is compact then $f(A)$ is also compact. But $f(A) \subseteq \mathbb{R}$ and hence $f(A)$ is bounded, i.e. f is bounded on A . ☆☆☆

5.1.6 REMARK. The requirement of the last theorem that A be compact is essential.

As for example, the function $f : (0, 1) \rightarrow \mathbb{R}$, $f(x) = 1/x$ is continuous, but $f((0, 1)) = (1, \infty)$. The conditions of theorem (5.1.5) are violated in this case since the domain $(0, 1)$ is not compact.

5.1.7 THEOREM. **(Weierstrass)**¹ If $f : (X, \mathcal{T}_X) \rightarrow \mathbb{R}$ is a continuous function and $A \subseteq X$ is a compact set, then f attains its greatest and least values.

¹Karl Weierstrass (1815-1897), a great German mathematician.

Proof. LIVE ☆☆☆

5.1.8 REMARK. *The requirement of the last theorem that A be compact is essential.*

For example, the function $f : (0, 1) \rightarrow \mathbb{R}$, $f(x) = x$, is continuous but $0 = m < f(x) < M = 1$ ($\forall x \in (0, 1)$).
The conditions of the above theorem are violated in this case since the interval $(0, 1)$ is not compact.

5.2 Continuous Mappings on Metric Spaces. Contractions

Let (X, ρ) and (Y, σ) be two metric spaces. The following definition follows directly from the definition of the continuity of a mapping on a topological space.

5.2.1 DEFINITION. A mapping $f : (X, \rho) \rightarrow (Y, \sigma)$ is said to be **continuous** at a point $x \in X$, if for any $\varepsilon > 0$, there exists $\delta(x, \varepsilon)$ such that $x' \in X$, $\rho(x, x') < \delta(x, \varepsilon)$ imply $\sigma(f(x), f(x')) < \varepsilon$.

5.2.2 DEFINITION. A mapping $f : (X, \rho) \rightarrow (Y, \sigma)$ is said to be **uniformly continuous** if for every $\varepsilon > 0$ there exists $\delta(\varepsilon)$ such that $x, x' \in X$, $\rho(x, x') < \delta(\varepsilon)$ implies $\sigma(f(x), f(x')) < \varepsilon$.

5.2.3 REMARK. *In order to prove that a mapping $f : (X, \rho) \rightarrow (Y, \sigma)$ is not uniformly continuous it is sufficient to find two sequences $(x_n), (y_n)$ such that,*

$$\delta(x_n, y_n) \rightarrow 0, \quad \text{and} \quad \sigma(f(x_n), f(y_n)) \not\rightarrow 0.$$

Note that a uniformly continuous mapping is a continuous mapping.

5.2.4 THEOREM. (**Cantor**) *If a function $f : (X, \rho) \rightarrow (Y, \sigma)$ is continuous and X is compact, then f is uniformly continuous.*

Proof. Let $\varepsilon > 0$. The continuity of f implies that for all $x \in X$ there exists $\delta(x, \varepsilon) > 0$ such that

$$\rho(x, x') < \delta(x, \varepsilon) \Rightarrow \sigma(f(x), f(x')) < \frac{\varepsilon}{2}.$$

The family

$$\left\{ B \left(x, \frac{\delta(x, \varepsilon)}{2} \right) \right\}_{x \in X},$$

is an open cover of X . Since X is compact then we can select a finite open sub-cover of X ,

$$\left\{ B \left(x_i, \frac{\delta(x_i, \varepsilon)}{2} \right) \right\}_{i=1, n}.$$

Denote:

$$\delta = \min \left\{ \frac{\delta(x_i, \varepsilon)}{2} \mid i = 1, \dots, n \right\}.$$

Let $x, x' \in X$ such that $\rho(x, x') < \delta$. From the fact that there exists $i \in \{1, \dots, n\}$ such that

$$x \in B \left(x_i, \frac{\delta(x_i, \varepsilon)}{2} \right)$$

we deduce:

$$\rho(x, x_i) < \frac{\delta(x_i, \varepsilon)}{2} < \delta(x_i, \varepsilon),$$

hence

$$\sigma(f(x), f(x_i)) < \frac{\varepsilon}{2}. \quad (\blacktriangleleft)$$

From the inequalities:

$$\begin{aligned} \rho(x', x_i) &\leq \rho(x', x) + \rho(x, x_i) \leq \\ &\leq \delta + \frac{\delta(x_i, \varepsilon)}{2} \leq \frac{\delta(x_i, \varepsilon)}{2} + \frac{\delta(x_i, \varepsilon)}{2} = \delta(x_i, \varepsilon), \end{aligned}$$

we deduce

$$\sigma(f(x'), f(x_i)) < \frac{\varepsilon}{2}. \quad (\blacktriangleleft\blacktriangleleft)$$

Taking into account the inequalities:

$$\sigma(f(x), f(x')) \leq \sigma(f(x), f(x_i)) + \sigma(f(x_i), f(x')) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

by $(\blacktriangleleft)$ and $(\blacktriangleleft\blacktriangleleft)$, we obtain

$$\sigma(f(x), f(x')) < \varepsilon.$$

The boxed text in the previous proof represents the definition of uniform continuity. ◆◆◆

5.2.5 REMARK. *The requirement of the last theorem that A be compact is essential.*

For example, the function $f : (0, 1) \rightarrow \mathbb{R}$, $f(x) = 1/x$, is continuous. Consider the sequences:

$$x_n = \frac{1}{n}, \quad y_n = \frac{1}{n+1}, \quad n \geq 1,$$

It can be seen that

$$x_n - y_n \rightarrow 0 \quad \text{and} \quad f(x_n) - f(y_n) = 1 \not\rightarrow 0.$$

The conditions of the above theorem are violated in this case since the interval $(0, 1)$ is not compact.

5.2.6 THEOREM. **(Heine)²** *A mapping $f : (X, \rho) \rightarrow (Y, \sigma)$ is continuous at a point $x \in X$, if and only if for any sequence $(x_n) \in X^{\mathbb{N}}$ converging to x , the sequence $(f(x_n))$ tends to $f(x)$.*

 no proof!

5.2.7 DEFINITION. A function $f : (X, \rho) \rightarrow (Y, \sigma)$ is called a **contraction** if there exists a number $\alpha \in [0, 1)$ satisfying the condition

$$\sigma(f(u), f(v)) \leq \alpha \rho(u, v),$$

for all $u, v \in X$.

5.2.8 REMARK. *It is obvious from the above definition that any contraction is a continuous function.*

5.2.9 DEFINITION. An element $x^* \in X$ is called a **fixed point** of a function $f : X \rightarrow X$ if

$$f(x^*) = x^*.$$

²Eduard Heine (1821-1881), a German mathematician.

5.2.10 THEOREM. (**Banach's Fixed Point Theorem**)³ If (X, ρ) is a complete metric space and $f : X \rightarrow X$ is a contraction then it has a unique fixed point.

Proof. LIVE

5.2.11 THEOREM. If I is a compact interval of the real axis and $f : I \rightarrow I$ is a function possessing derivative such that $|f'(x)| \leq \alpha < 1$, for all $x \in I$, then the equation $f(x) = x$ has a unique root which can be approximated by the fixed-point iteration method.

Proof. LIVE

5.2.12 EXAMPLE. Approximate the root of the equation

$$x^5 + 7x - 7 = 0$$

in the interval $[0, 1]$.

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5.2.13 EXAMPLE. Approximate the solutions of the system of equations

$$\begin{cases} \sin y = 2x \\ \cos x = 2y \end{cases} \quad (x, y) \in \mathbb{R},$$

using the fixed-point iteration method.

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5.3 Continuous Mappings on Euclidean Spaces

In this section we study the concept of continuity for Euclidean space $\mathbb{R}^n$. We recall that, for the Euclidean space $\mathbb{R}^n$, the metric ρ is given by

$$\rho((x_1, \dots, x_n), (y_1, \dots, y_n)) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}.$$

Consider the Euclidean spaces $\mathbb{R}^n$ and $\mathbb{R}^m$ and a mapping

$$f : \mathbb{R}^n \rightarrow \mathbb{R}^m.$$

There exist the functions

$$f_i : \mathbb{R}^n \rightarrow \mathbb{R}, \quad i = 1, \dots, m,$$

such that

$$f = (f_1, \dots, f_m),$$

i.e.

$$f(x_1, \dots, x_n) = (f_1(x_1, \dots, x_n), \dots, f_m(x_1, \dots, x_n)).$$

We denote:

$$x = (x_1, \dots, x_n), \quad y = (y_1, \dots, y_n) \in \mathbb{R}^n.$$

From the definition of the continuity of a mapping on a metric space, we deduce:

A mapping $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is continuous at the point $(y_1, \dots, y_n) \in \mathbb{R}^n$ if for any $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\text{if } \sqrt{\sum_{i=1}^n (x_i - y_i)^2} < \delta \quad \text{then} \quad \sqrt{\sum_{j=1}^m (f_j(x) - f_j(y))^2} < \varepsilon.$$

³Stephan Banach (1892-1945), a Polish mathematician.

We deduce that the function f is continuous if and only if its component functions $f_1, \dots, f_m$ are continuous.

The limit of the mapping f at a point is defined as follows:

$$\lim_{x \rightarrow y} f(x) = \ell = (\ell_1, \dots, \ell_m) \in \mathbb{R}^m$$

if for any $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\text{if } \sqrt{\sum_{i=1}^n (x_i - y_i)^2} < \delta \quad \text{then} \quad \sqrt{\sum_{j=1}^m (f_j(x) - \ell_j)^2} < \varepsilon.$$

5.3.1 EXAMPLE. Investigate the continuity of the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$

$$f(x, y) = \begin{cases} \frac{x^2 + y^2}{|x| + |y|}, & \text{if } (x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\} \\ 0, & \text{if } (x, y) = (0, 0) \end{cases}$$

at the point $(0, 0)$.

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5.3.2 EXAMPLE. Examine the continuity of the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2}, & \text{if } (x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\} \\ 0, & \text{if } (x, y) = (0, 0) \end{cases}$$

at the point $(0, 0)$.

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5.3.3 EXAMPLE. Investigate the continuity of the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$

$$f(x, y) = \begin{cases} (x, \sqrt{|xy|} \sin \frac{1}{\sqrt{|xy|}}), & \text{if } xy \neq 0, \\ (x, 0), & \text{if } xy = 0. \end{cases}$$

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